

```
9. sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
10. plt.title('Correlation Matrix of Tech Stock Returns (2022)')
11. plt.show()
12.
```

This heatmap instantly reveals:

- Which stocks move most similarly (highest positive correlation)
- Which stocks might provide diversification (low or negative correlation)
- Potential pairs for statistical arbitrage (highly correlated pairs)

Correlation analysis can uncover non-obvious relationships in the market. For example, you might discover that a particular stock correlates strongly with a commodity price or currency movement, providing predictive signals for your trading.

Time Series Analysis: Understanding Market Behavior Over Time

Financial data is fundamentally time series data—observations collected sequentially over time. This special structure requires specialized analytical techniques.

Stationarity and Non-Stationarity: The Foundation of Time Series Analysis

A key concept in time series analysis is stationarity. A stationary time series has statistical properties that don't change over time—the mean, variance, and autocorrelation structure remain constant. Most raw financial price series are non-stationary (they trend and vary in volatility), while return series are often closer to stationary.

This distinction is crucial because many analytical techniques require stationary data to work properly. Let's test whether Apple's price and returns